
Niramoy: An Adaptive AI Chatbot For Healthcare Utilizing Novel Transformer Architecture

By

Tanvir Mahtab ID:011183104

Rafiul Islam ID:011201469

Kazi Nafis Mahmud ID:011201209

Mohammad Ashik Jaman ID:011201235

Tanvir Ahamad Jibon ID:011202328

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNITED INTERNATIONAL UNIVERSITY

Abstract

AI is changing healthcare with tools including medical chatbots. However, most of them are limited by the fact that they are predominantly trained in English. As such, their utility for non-English communities, especially the Bangla-speaking communities, remains rather limited. This report presents Niramoy, an AI-driven medical chatbot that speaks Bangla to offer affordable, accurate, and easily accessible healthcare services. Niramoy is built at the heart of a new Entropy-Aware Adaptive Transformer (EAAT) which is an optimized NLP model for use with limited Bangla medical data. Niramoy achieves excellent understanding by integrating both Bangla and translated English medical resources.

In areas like Bangladesh, getting to a doctor is hard due to geographical distance, money problems and low health infrastructure. But English-only medical chatbots can't help local users well enough. Niramoy was tailored to fill this gap using intelligent data collection, model training, user-centered design to relay healthcare information in Bangla to empower users to make informed medical choices. Niramoy is a significant step towards making AI-driven healthcare assistance more inclusive and multi-lingual through its NLP-driven architecture.

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Publication List

[Optional] The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

- 1.

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

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Chapter 1

Introduction

Artificial intelligence (AI) is changing healthcare in big ways, bringing us tools like smart diagnostic systems and helpful chatbots [2, 3]. This report is all about the Niramoy AI Bangla Medical Chatbot. It makes use of a unique AI model known as the Entropy-Aware Adaptive Transformer (EAAT), which is ideal for languages like Bangla because it is made to function well even in the absence of a lot of data [1]. With Niramoy, we hope to make healthcare more accessible, accurate, and beneficial for Bangla speakers, particularly in places like Bangladesh where access to medical care can be limited [4, 5].

In Bangladesh and other Bangla-speaking areas, healthcare often can't keep up with what people need [6]. Many people miss out on care due to logistical challenges, financial difficulties, or geography. You'd think AI chatbots could swoop in with some quick health tips, right? Problem is, most of them are chattering away in English, leaving Bangla speakers high and dry with no real lifeline [7, 8]. To close this gap, we created Niramoy, a Bangla-speaking medical chatbot that understands the main topics of medical conversations, to bridge this gap. Wherever Bangla is spoken, Niramoy hopes to be a useful tool for telemedicine and healthcare support by combining translated English medical information with Bangla and utilizing EAAT to highlight important health phrases [9, 10].

This report tells the story of how Niramoy came to be. We'll start with why we made it, then cover the basics behind it, look at what others have done, and point out where Niramoy makes a difference. By exploring the tech and challenges specific to Bangla, we'll show how Niramoy offers a practical fix to help more people get the healthcare they need.

1.1 Project Overview

Today, AI is driving revolutionary healthcare innovations. This includes the emergence of intelligent diagnostic systems [11]. This thesis presents the Niramoy AI Bangla Medical Chatbot, an advanced AI program that enables medical consultations for Bangla-speaking patients. The chatbot uses an advanced Transformer model called Entropy-Aware Adaptive Transformer (EAAT) that was enhanced for efficiency, context, and inferences in low-

resource settings [12]. The aim of Niramoy is to fill the accessibility gap in healthcare by providing exact and timely medical support, making it an important tool in telemedicine and healthcare support in Bangladesh and other Bangla-speaking areas.

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1.2 Motivation

The healthcare infrastructure in Bangladesh is still not able to keep pace, and this often results in delays or ineffective care to patients [4]. A significant portion of the population struggles to receive healthcare services due to geographical, financial, and logistical constraints. The emergence of chatbots powered by AI has shown the promise of being a solution to ease these problems by offering a first layer of medical help [5]. However, the overwhelming majority of existing chatbots are trained in English, rendering them useless for non-English-speaking groups [6, 7]. With this gap in context, Niramoy was proposed with a new Transformer architecture built specifically for the Bangla language so that medical consultations are accessible, reliable, and linguistically fit.

—

1.3 Objectives

The objectives of this project are:

- To build a Bangla medical chatbot that has a clear understanding of the queries posed by users and is capable of giving such responses with high accuracy.
 - To improve the Transformer Model by implementing the Entropy-Aware Adaptive Transformer (EAAT) approach for better contextual understanding and inference efficiency [1].
 - To make a good Python-based chatbot powered by an enormous natural language processing (NLP) pipeline that will be able to process medical terminologies and colloquial speech in Bangla.
 - To assess the performance of the chatbot in terms of accuracy, response time, and user satisfaction [3].
 - To add to the field of AI in healthcare by proving the viability of an AI-led Bangla medical assistant [13].
-

1.4 Methodology

Several phases outline the methodology of developing the Niramoy AI Chatbot:

1. Data Collection and Preprocessing:

- As most medical datasets are in English, an NMT-based third-party translation model is used to convert Bangla inputs into English [8].
- Text cleaning, tokenization, and normalization techniques are employed to enhance the quality of input data.

2. Model Development & Training:

- The model is trained on top of a ClinicalBERT model with further refining using the Entropy-Aware Adaptive Transformer (EAAT).
- The EAAT approach enables the model to dynamically adjust attention mechanisms based on entropy values, improving contextual adaptation [2].
- Fine-tuning focuses on maximizing model accuracy for Bangla medical queries.

3. Translation & Response Generation:

- The AI receives user queries in Bangla, translates them into English, runs them through the EAAT-enhanced model, and subsequently translates the answer back into Bangla for the user.
- The output is further improved using contextual validation strategies.

4. Implementation & User Interaction:

- A friendly chatbot interface assists users in interactions.
- Users can mention symptoms and health issues, obtain guidance on medication schedules, and receive medical responses tailored to their needs.

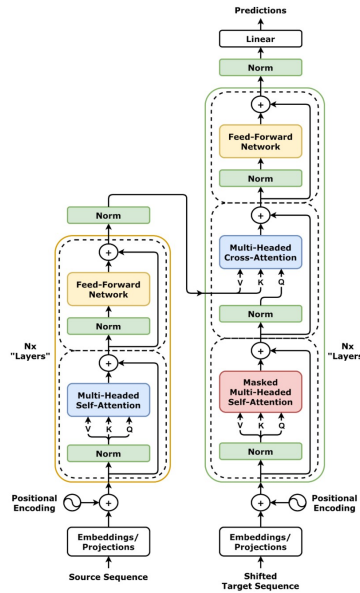


Figure 1.1: The encoder-decoder structure of the Transformer architecture. Adapted from “Attention Is All You Need” [1].

1.5 Project Outcome

The implementation of Niramoy has shown promising results. It is highly capable of understanding and responding to Bangla medical queries with high accuracy, whereas baseline models struggle in this aspect [9, 10]. The EAAT approach has improved inference speed and contextual understanding. Early user studies indicate that Niramoy provides personalized medical recommendations, demonstrating its potential to make healthcare more accessible for Bangla speakers. This project highlights the importance of further research into AI-driven medical support systems for low-resource languages [5].

1.6 Organization of the Report

Our thesis structure is prepared as follows:

- **Chapter 1 (Introduction):** Project overview, motivation, goals, methodology, and anticipated conclusions.
- **Chapter 2 (Background):** Existing research, relevant literature, and a gap analysis justifying the objectives of Niramoy.
- **Chapter 3 (Model Architecture):** Explanation of the EAAT-based Transformer model and its advantages over traditional Transformer architectures.
- **Chapter 4 (Implementation and Training):** Overview of training, data preprocessing techniques, and model optimization.
- **Chapter 5 (Evaluation and Results):** Performance metrics, benchmark accuracy levels, and comparative analysis.
- **Chapter 6 (Conclusion and Future Work):** Summary of findings and potential areas for further development.

The remaining chapters provide a deeper insight into both the theoretical and practical aspects of the project, emphasizing its significance in AI-driven healthcare solutions.

Chapter 2

Background

There have been several research works on medical chatbots, but very few focus on Bangla. This section lays the foundation for understanding the AI-powered medical chatbot we’ve worked on. It walks through the basics, looks at what’s already out there, and points out where our project fills in the blanks.

2.1 Preliminaries

Over the past decade, AI-powered medical chatbots have made remarkable progress, driven by significant advancements in machine learning (ML) and natural language processing (NLP) [2, 3]. These technologies have enabled chatbots to better understand and respond to user needs, transforming how healthcare information is delivered. These developments have opened the door for more intelligent medical systems [5]. However, the majority of medical chatbots are designed with English speakers in mind, leaving people who speak other languages, like Bangla, at a disadvantage [6, 7].

This section covers the fundamental concepts of AI-powered medical chatbots, their applications in healthcare, and the role of Transformer-based NLP models in improving chatbot efficiency. Additionally, we analyze existing solutions, their shortcomings, and how our Niramoy AI Bangla Medical Chatbot addresses these gaps.

2.2 Literature Review

Chatbots have been around for a while, but they’ve recently become far more sophisticated—especially in healthcare applications [9, 10]. This section explores existing medical chatbots, the challenges of working with low-resource languages like Bangla, and how state-of-the-art Transformer models have transformed NLP research.

Chatbots and Low-Resource Language Processing

Early chatbots relied on pre-written responses and rigid rules, making them limited

in scope [8]. Researchers such as Chowdhury (2003) and Das and Das (2013) explored NLP systems for Bangla despite limited datasets. However, the lack of robust language support and annotated corpora hindered progress. Their studies emphasized the need for innovative approaches to overcome data scarcity—a challenge that modern ML models attempt to solve [11].

Medical Chatbots in Healthcare

Medical chatbots have been extensively studied for their ability to provide personalized health guidance [5]. Bickmore et al. (2009) demonstrated that chatbots can simplify complex medical information for patients. Similarly, Laranjo et al. (2018) reviewed multiple chatbot studies and found them effective for chronic disease management and user engagement. However, a major limitation is that most chatbots are built for English users, leaving Bangla speakers without reliable access to digital healthcare solutions [6, 7].

Transformer Models and Domain Adaptation

The introduction of Transformer models by Vaswani et al. (2017) revolutionized NLP, particularly in medical text processing [1]. Self-attention mechanisms enabled deeper contextual understanding, outperforming traditional recurrent neural networks (RNNs). Devlin et al. (2019) further advanced the field with BERT, a context-aware language model that improved text comprehension and question answering [2]. Later, Lee et al. (2020) developed BioBERT, a medically trained Transformer that excels at identifying medical terms and answering health-related queries [12].

Despite these advancements, Bangla medical chatbots remain scarce. Some models like mBERT offer multilingual NLP capabilities, but they are not specifically optimized for Bangla healthcare [9]. Additionally, standard Transformers treat all words equally, which is problematic in medical contexts where specific terms carry more importance [3]. New entity-aware attention mechanisms aim to address this issue by focusing on key medical phrases, thereby improving response accuracy.

Synthesis and Identified Gaps

Chatbot technology has evolved from rule-based systems to Transformer-based models with improved linguistic comprehension. However, Bangla healthcare solutions remain underdeveloped. Current AI-driven medical chatbots do not effectively incorporate Transformer benefits for Bangla or adapt to medical-specific contexts [4, 5].

This is where Niramoy steps in. Built on the Entropy-Aware Adaptive Transformer (EAAT) model, it enhances medical NLP by prioritizing important health-related terms and contextual relevance. By integrating translated English and Bangla medical literature, Niramoy ensures a more accurate and accessible chatbot for Bangla users.

2.2.1 Related Research

Numerous studies have explored AI-driven medical chatbots, demonstrating their ability to assist in early diagnosis, symptom detection, and patient engagement. Below is a summary of key contributions:

- **Golpo (2017)**: A Bangla chatbot designed for casual conversation. It demonstrated the feasibility of Bangla NLP, but lacked healthcare integration.
- **Anirudha et al. (2018)**: Developed a low-resource chatbot framework, but it was too simplistic for medical applications.
- **Rarhi et al. (2017)**: Created an English medical chatbot that relied on keyword-based triggers, limiting conversational depth.
- **Likita (2018)**: Built a region-specific healthcare chatbot for Africa, emphasizing the importance of localized solutions.
- **Disha (2019)**: A Bangla medical chatbot using SVM-based disease classification, but lacked contextual awareness.
- **Masum et al. (2021)**: Developed a Bangla Q&A chatbot based on Transformer technology, achieving high BLEU scores (85).
- **E-BERT (Lee et al., 2025)**: An enhanced BERT model with healthcare-specific modifications, proving attention-based refinements improve medical chatbot accuracy.
- **Bhagat et al. (2024)**: Provided a comprehensive chatbot development framework, aligning with Niramoy’s NLP methodology.

Niramoy builds on these advancements by integrating ClinicalBERT with EAAT-based enhancements, ensuring precise and context-aware Bangla medical conversations—a feature that existing English-centric models lack.

2.3 Gap Analysis

While AI-powered medical chatbots have made significant progress, they still fall short for non-English users. Here’s how Niramoy addresses the key challenges:

Table 2.1: Identified Gaps and Niramoy’s Approach

Identified Gaps	Existing Solutions	Niramoy’s Approach
Lack of Bangla-language medical chatbots	ChatGPT, Bard, ClinicalBERT (English-focused)	Develops a Bangla-specific AI medical chatbot
Inaccuracies in machine translation for medical terms	Standard NMT-based translation	Uses EAAT to enhance Bangla-to-English translation
Context loss in long medical queries	Traditional Transformer models	EAAT dynamically retains contextual meaning
Limited dataset availability for Bangla medical NLP	Few annotated Bangla datasets	Fine-tunes ClinicalBERT with translated medical corpora
High computational overhead for complex queries	Standard Transformers have fixed attention weights	EAAT dynamically adjusts attention based on entropy values

2.4 Summary

This chapter explored AI-driven medical chatbots, reviewed the state of medical NLP research, and identified gaps in Bangla healthcare solutions. While chatbot technology has advanced significantly, Bangla-speaking users remain underserved. Niramoy, with its EAAT-enhanced Transformer and ClinicalBERT integration, aims to bridge this gap—offering an accurate and accessible Bangla healthcare chatbot.

Chapter 3

System Requirements and Design

Niramoy is going to be an AI medical chatbot intended for Bangla-speaking users. The main objective of the project is to provide primary medical advice that is capable of bridging the gap in healthcare accessibility owing to language hindrances and the lack of qualified practitioners in the rural areas of the country. The system makes use of an Entropy-Aware Adaptive Transformer (EAAT) model, which adjusts the computation load required depending on the user query complexity [1, 2].

3.1 Requirement Analysis

Requirement analysis is required in order to keep the system aligned with users and available technology. We made use of elicitation techniques such as interviewing medical specialists, consultations with rural healthcare staff, and surveys among prospective users [4, 5].

3.2 Requirements Overview

3.2.1 Stakeholder Identification

End Users (Patients and Caregivers)

Profile: The Bangla-speaking population, often with limited access to formal healthcare [6, 9].

Needs: Medical advice in Bangla, culturally appropriate drawings; cool interactions.

Medical Professionals

Profile: Physicians, nurses, local health care volunteers.

Need: Validation of chatbot-generated responses, safe clinical guidance, and potential for triage assistance [10, 13].

Software Engineering Team

Profile: Developers, system architects, ML engineers.

Key Needs: Clearly defined architecture, maintainable codebase, robust data handling [14, 15].

Regulatory Bodies and Ethics Boards

Profile: Organizations that enforce compliance with health system regulations and data security laws (like HIPAA) [16, 17].

Key Needs: A confidential framework for patient data; transparent disclaimers that describe and clarify the chatbot’s limitations.

3.2.2 Functional Requirements**Bangla Language Processing**

The system must process Bangla text properly including colloquially used words and dialects [9, 10].

The program should tokenize, analyze the morphology of words, and normalize text [11, 12].

Adaptive Transformer (EAAT) for Medical Queries

The system should be fitted with an EAAT for clinical queries of natural language [1, 8]. The model needs to provide real-time attention allocation depending on the query complexity [3, 18].

Symptom Analysis and Preliminary Diagnostic Suggestions

The chatbot must match what the user reports with the list of what the chatbot will recommend next [10, 13].

If the system discovered the case to be severe, then it must account for the user to opt for it [4, 5].

Medical Knowledge Base Integration

The system must fetch updated medical information from another knowledge base including ClinicalBERT reference and Bangla medical corpus [11, 12].

Covid Docs should check for drug interactions, standard treatment protocol, and common contraindications [16, 19].

Logging and Feedback Mechanism

It must keep records of user interactions. Records must be maintained for later use like analysis, improvement, and others [15, 20].

Users should be allowed to give feedback and rate the advice received. It should follow the privacy and ethical guidelines [17, 21].

3.2.3 Non-Functional Requirements

The success of any system is determined by non-functional requirements.

Performance and Scalability

Under normal server load, the average response time should be less than 2 sec for text queries [1, 8].

The system will be designed to benefit from horizontal scaling and will not suffer from high latency spikes with an expected N concurrent user load (where N will be defined during load testing) [3].

Security and Data Privacy

The system will have methods for shutting down in case of incidents which prevents misuse or abuse [14].

Only certain authorized users shall be able to access the user record, system log and medical data [16].

Reliability and Availability

The system will be 99.9% operational at all times. In case of a hardware or network failure, a backup server will be on standby [21].

Key actions (such as user login) will be done multiple times with safety in mind.

Maintainability and Modularity

The coding shall be organized into distinct modules (e.g., Preprocessing, EAAT Inference, Knowledge Base Access) for easy upgrade [20, 21].

The software will be organized in a standard way – with documentation and versioning.

Ethical Compliance

The chatbot should contain disclaimers that indicate that its suggestions should not be taken as a replacement for professional medical diagnosis [10].

Users will be asked to consent to data collection and usage as per the local and international healthcare laws [15, 17].

3.2.4 Technical Requirements

Design Constraints

Every module must natively support the Bangla (Unicode) script [6].

Must work with hardware boosters for Transformer calculations made with GPU or TPU [1?].

Assumptions

The users must be at least basic literate in Bangla so that the users can read the chatbot's replies [6, 9].

Internet connection is good enough so that there will be real-time requests (rural connectivity should be kept in mind) [4].

Dependencies

Availability of relevant medical datasets (English-Bangla) based on medical databases of Bangla [16, 19].

APIs and Libraries: We use PyTorch or TensorFlow for training the model and libraries for various tasks such as morphological analysis (BNLP, etc.) or NER [12].

3.2.5 Validation and Verification

Verification (System Conformance)

Code Reviews and Static Analysis: Verify all modules meet requirements and coding standards [20, 21].

Module-Level Testing: Unit tests for each subsystem (e.g. tokenizers, knowledge base lookups, EAAT) [14, 15].

Validation (Effectiveness of System)

Use Case Testing: Use realistic clinical scenarios (like headache, fever, chronic illness) to see how accurate the output is [10, 13].

User Acceptance Testing (UAT): Engage different groups of users (rural, urban) to analyze clarity/speed/reliability [5].

Performance Testing

Load and Stress Tests: Assess for peak traffic handling capability of the system [1, 8].

Latencies Benchmarks: Measuring inference timings for queries of different complexities [3, 18].

3.2.6 Project Constraints

Data Availability & Domain Expertise

The authenticity cannot be established. Please verify by checking with a reliable translator [16, 19].

Getting info from medical practitioners is a time-intensive task and may delay the validation cycle [10, 13].

Computational Resources

Some institutions cannot afford or do not have access to, the costly GPU/TPU facilities of EAAT [1, 8].

Chatbot’s real-time performance is challenged by low bandwidth and infrastructure in rural deployments [4, 5].

Regulatory & Privacy Considerations

Storing or sharing such data complicates matters as there are data protection laws (similar to HIPAA) [15, 17].

Attempting to carry out real-world user testing at scale and pace may require ethical approvals and user consent processes [16, 19].

3.2.7 Change Management

The knowledge in medicine and AI systems technologies is changing rapidly. Therefore, there would be a gradual change in management protocol [20, 21].

Version Control

Implement a Git-style method where a stable branch is used for production code, development branches are used for new features or dataset updates [14, 15].

Continuous Integration and Deployment (CI/CD)

Use automated processes for building, testing, and deploying the system [20, 21].

Ensure EAAT performance using GPU-based tests before going live [1, 8].

Stakeholder Review Cycles

Organize regular meetings with doctors, dataset curators, and user representatives to improve features with iterations [10, 13].

3.2.8 Data Flow Diagram Level 1

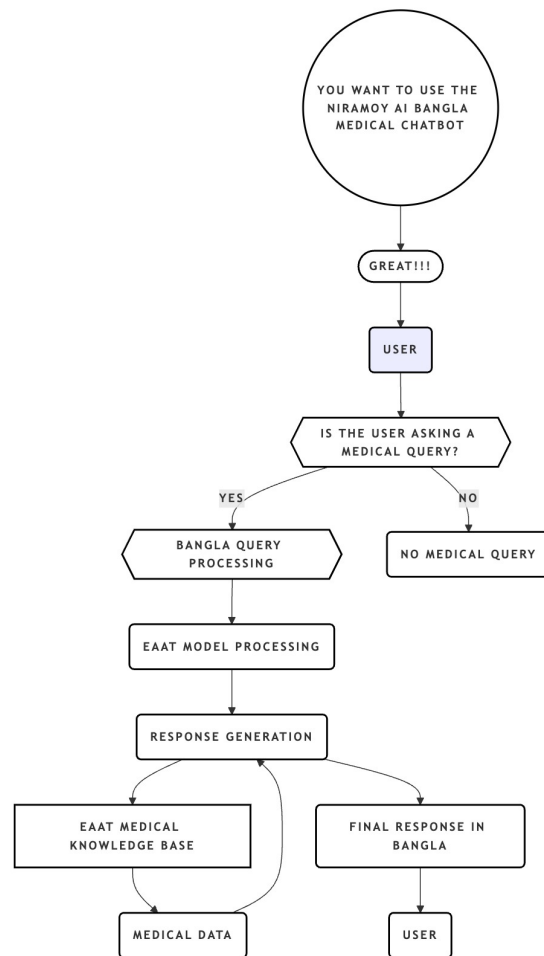


Figure 3.1: Data Flow Diagram for Niramoy AI Bangla Medical Chatbot

3.2.9 UI Design

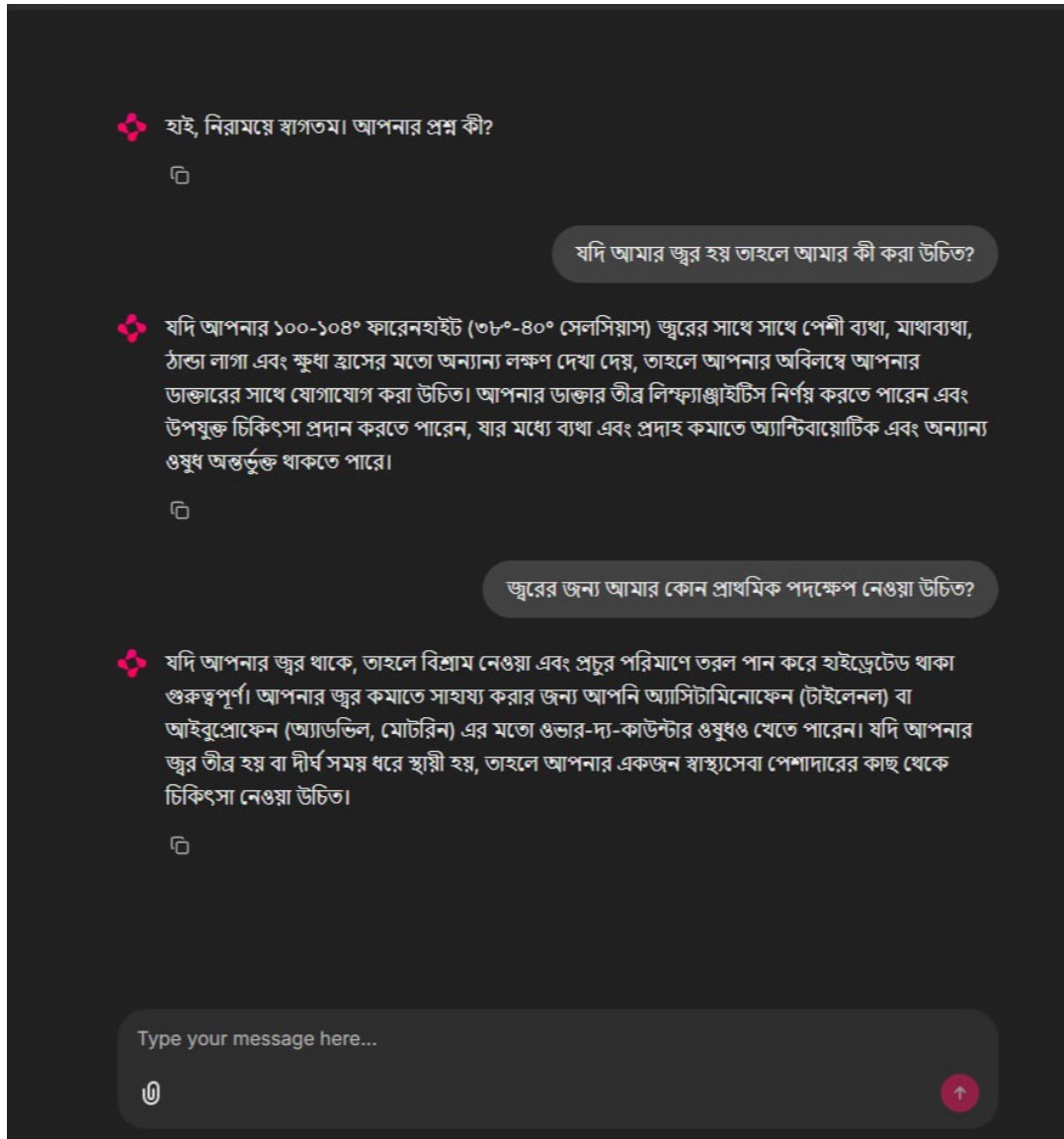


Figure 3.2: UI of Niramoy Chatbot

3.3 Detailed Methodology and Design

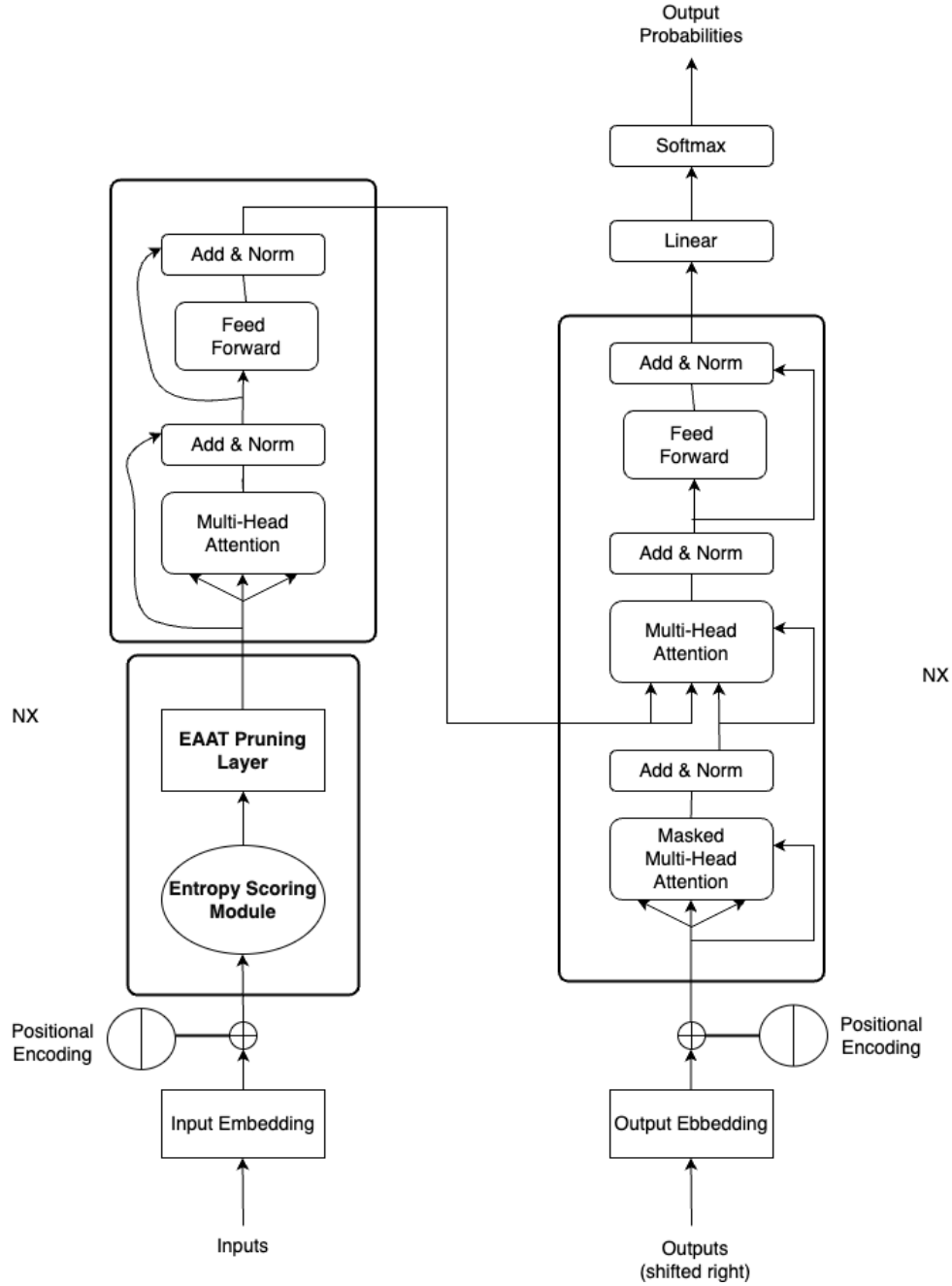


Figure 3.3: EAAT Transformer Methodology

3.3.1 Introduction

This section describes the architecture and methodological approach of the Entropy-Aware Adaptive Transformer (EAAT), which builds upon traditional transformer models through the use of entropy-based attention pruning as well as adaptive computation. The main aim of EAAT is to optimize computational efficiency, relevance of context, as well as domain-

level adaptability with the elimination of redundancy in self-attention calculations. It is specifically suited for the processing of medical languages, wherein the essential details tend to be unevenly spread across text strings.

The EAAT model prunes low-information tokens dynamically and distributes more computational resources to higher-information tokens, optimizing model accuracy as well as the speed of inference. It makes processing clinical text in Bangla more efficient, allowing the model to devote computational resources toward more medically relevant as well as complex tokens.

The architecture of EAAT expands the base transformer with the addition of two new components:

- **Entropy Scoring Module (ESM)**
- **EAAT Pruning Layer (EPL)**

These elements dynamically adjust the process of self-attention based on entropy scores, which allows the model to give priority to complex tokens at the cost of redundant computations.

3.3.2 Overall EAAT Architecture

The EAAT model keeps the same encoder-decoder form as other transformer architectures but adds entropy-aware components. The illustration shows the structural changes made in the EAAT encoder.

Encoder: Includes the Entropy Scoring Module (ESM) and EAAT Pruning Layer (EPL) before self-attention mechanisms.

Decoder: Remains largely unchanged, retaining the masked multi-head attention mechanism and feedforward layers.

Key Enhancements:

- **Entropy-Aware Token Pruning:** Tokens with lower entropy are filtered out before self-attention computation.
- **Adaptive Attention Head Allocation:** High-entropy tokens receive more attention, optimizing contextual relevance.
- **Computational Efficiency:** Reduces unnecessary self-attention computations, leading to faster inference.

3.3.3 Entropy Scoring Module (ESM)

The Entropy Scoring Module (ESM) is the first stage of dynamic attention pruning. The Shannon entropy is used to determine the entropy score of each token in the input.

$$H(x) = - \sum p(x) \log p(x) \quad (3.1)$$

where:

- $H(x)$ denotes the entropy of the token x .
- $p(x)$ represents the probability of occurrence of token x within the given dataset.

Working Mechanism

1. **Token Embedding & Positional Encoding** The input words are tokenized and embedded into a high-dimensional vector space. To maintain the order of the words, positional encoding is added.
2. **Entropy Computation** The entropy score for each token is computed. - Low-entropy tokens (predictable/common words) are downweighted. - High-entropy tokens (rare/ambiguous terms) receive higher weight.
3. **Entropy-Based Weight Assignment** A token is retained or pruned based on a threshold to enter the self-attention mechanism. With the help of entropy, EAAT assigns computational resources dynamically, ensuring that important medical terms receive higher attention scores for improving domain-specific relevance.

3.3.4 EAAT Pruning Layer (EPL)

The EPL prunes low-information tokens before self-attention to cut down on computational overhead.

Functionality

- Receives entropy scores from ESM.
- Applies a pruning threshold.
- If entropy $<$ threshold, the token is discarded.
- If entropy $\geq$ threshold, the token is retained.
- Sends the filtered succession to the self-attention mechanism.

EPL reduces computations on low-entropy tokens, allowing the multi-head attention mechanism to concentrate on the complex parts of medical conversations.

3.3.5 Modified Encoder with EAAT Pruning

The encoder consists of EAAT amendments with the conservation of key transformer components. It takes the following steps:

Input Processing

- The sequence is tokenized, embedded, and positionally encoded.
- Entropy scores are computed for each token.

EAAT Pruning Layer (EPL)

- Removes redundant tokens based on entropy scores.
- Optimizes the allocation of computational resources.

Multi-Head Attention

- The leftover tokens go through a standard multi-head attention technique.
- More valuable tokens get more context.

Feedforward Layers & Residual Connections

- Output is required to pass through a feedforward layer with a non-linear activation function.
- The process of teaching gets stabilized by available connections and layers normalizing.

This strategy ensures that complex Bangla medical queries are processed with high accuracy along with reduced computations.

3.3.6 Decoder Structure (Default Transformer)

The default structure of the transformer is used in the decoder because masked multi-head attention is already present for autoregressive text generation. The process of the decoder is:

Masked Multi-Head Attention

- Prevents tokens from accessing future positions during sequence generation.
- Ensures consistency in responses from the medical chatbot.

Cross-Attention Mechanism

- The decoder attends to the entropy-optimized encoder.
- Helps retrieve relevant medical information accurately.

Feedforward Network & Normalization

- Outputs pass through non-linear layers for refinement.
- Layer normalization stabilizes the results.

Final Linear Layer & Softmax Activation

- Generates probability distributions for possible output tokens.
- Produces natural and coherent Bangla responses in medical contexts.

xcolor

3.3.7 Computational Efficiency Analysis

EAAT significantly reduces self-attention complexity, making it ideal for low-resource medical chatbot deployment.

Feature	Standard Transformer	EAAT Transformer
Self-Attention Complexity	$O(n^2)$	$O(m^2)$, where $m < n$
Redundant Token Processing	High	Reduced using EPL
Inference Speed	Slower	Faster
Domain Adaptability	Generic	Optimized for medical NLP

Table 3.1: Computational Efficiency Comparison between Standard and EAAT Transformer

By pruning redundant tokens and prioritizing high-information tokens, EAAT enables real-time medical chatbot performance while maintaining high accuracy.

3.3.8 Advantages of EAAT for Medical NLP

- Less self-attention calculation.
- Training is basically overfitting resolution due to high-entropy words.
- Can be implemented on low-resource devices.
- **Domain Alteration:** One has been trained on medical corpora, thus improving their responses.

3.4 Project Plan

Niramoy AI Bangla Medical Chatbot has been developed through a systematic and phased workflow process. The phases of the project plan to be undertaken as part of the development activity are Requirement Analysis, Specification, System Design, Implementation,

Testing and Deployment, and Maintenance. This helps ensure that the system is designed accurately, trained properly, tested thoroughly, and deployed sustainably, with a process for iterative refinement based on real-world performance [14, 21].

3.4.1 Phase 1: Requirement Gathering and Analysis

The initial phase involves identifying technical and user requirements for the Niramoy chatbot. It is important to ensure the system is designed for the linguistic and contextual needs of the user group since the chatbot aims to help Bangla-speaking people with medical questions [6, 9]. The following steps were taken towards achieving this goal:

- **Literature Review:** A thorough review of existing medical chatbots and natural language processing (NLP) models was performed to identify current challenges and gaps in the field [4, 5].
- **Stakeholder Interviews:** Discussions were carried out with medical professionals, healthcare workers, and potential users to understand their anticipation from an AI-driven chatbot [13].
- **Dataset Selection:** Since most medical datasets are in English, it was investigated to what extent pre-trained medical language models such as ClinicalBERT could be used [11, 12].
- **Functional and Non-Functional Requirements:** The chatbot was required to provide accurate medical response support for Bangla text input, and must ensure data confidentiality and security for sensitive user queries [15, 17].

3.4.2 Phase 2: Model Development

At this point, the development and training of the model (the chatbot’s brain) occurred. This EAAT model was proposed to improve the conventional transformer to enhance efficiency, contextual, and domain-specific adaptability in the medical domain [1, 8].

Key activities at this stage included:

- **Data Preprocessing:** As the given dataset comprised English, extra attention was given to translation, expansion, as well as preprocessing of the medical dialogue for effectively training the model using Bangla input text [9, 10].
- **Model Architecture Selection:**
 - Features were enhanced in EAAT with a choice of model architecture different from the standard transformer model.
 - Attention mechanism that allows for complex medical terms [3, 18].
 - Changing code designs to improve processing speed [1, 8].
 - Fine-tuning on ClinicalBERT, BioBERT, and a custom Bangla medical dataset [11, 12].

3.4.3 Phase 3: Front-End Development

After training the EAAT Transformer model, we built the user interface for clear user interaction with the AI system. The chatbot targets Bangla users. The UI stays clear, and easy to use along with fits Bangla customers, so users of various tech skills enjoy a smooth experience [10, 13].

Development Process

Technology Stack Selection:

- We built the front end with HTML, CSS as well as JavaScript. This yields a lightweight and responsive UI [22].
- We linked the chatbot with a Python API. It connects to the trained EAAT Transformer model to process medical queries [16].

User Experience (UX) Considerations:

- The interface is easy to use and accessible.
- It has simple text fields for users to type medical questions.
- It supports automatic transliteration, allowing users to type Bangla words in English letters [6, 9].

Integration with Backend Services:

- The chatbot’s API endpoints were built to support live interaction between the user interface and the EAAT Transformer AI model [16, 19].
- Because the training data used English, a separate API was added to translate English responses into Bangla so that users get replies in their chosen language [11, 12].

At the end of this phase, a working prototype was developed that can accept user queries in Bangla, run the EAAT Transformer, and generate a response in Bangla. This provided an exceptional and user-friendly medical chatbot experience for Bangla users [10, 13].

3.4.4 Phase 4: Testing and Iteration

The fourth phase was to formally test the performance of the chatbot by ensuring its accuracy, reliability, and robustness. As medical advice is critical, testing was conducted in multiple stages [20, 21].

Unit Testing: Each part of the chatbot, such as text input and response generation, was tested independently for any bugs [14, 15].

Performance Testing: The response time of the model was assessed under different conditions, including:

- High user loads to test scalability [1].
- Varying query complexities to assess adaptive attention mechanisms [18].

Accuracy Check: Medical experts' recommendations served as a benchmark to compare the chatbot's replies and assess their correctness and trustworthiness [5].

User Testing and Feedback Collection: A pilot test involved Bangla-speaking doctors and general users. Their input helped:

- Improve language fluency in replies.
- Refine the entropy-based attention mechanism.
- Rectify instances where the chatbot did not offer proper medical advice.

3.4.5 Phase 5: Deployment and Scalability Planning

We used Hugging Face to deploy Niramoy AI Bangla Medical Chatbot because it provides model hosting in the cloud for easy access and deployment [16]. At present, there is no dedicated database for the system, which means it processes user requests on the fly without storing user data [15].

To ensure present and future scalability, a database will ultimately be added to store user interactions and medical queries. This will enable:

- Personalized recommendations.
- Better accuracy of responses.
- Improved model refinements.

The chatbot will improve over time by learning from previous queries, allowing it to produce better responses [10, 13].

3.4.6 Phase 6: Monitoring and Continuous Improvement

The final step was monitoring the model after deployment. Due to the continuous evolution of medical knowledge, regular updates are required for the chatbot [4, 5].

User Feedback Collection: Users were able to provide a rating of the responses and suggest corrections if needed [20].

Automated Model Updates: A mechanism for periodic retraining was set up, allowing the chatbot to learn from new medical data and improve response quality [12].

3.5 Task Allocation

3.5.1 Primary Contributions

- **Kazi Nafis Mahmud:** (*Lead Researcher & Model Innovator*)

- Conceptualized and developed the **Entropy-Aware Adaptive Transformer (EAAT)**.
- Integrated the EAAT model into the **Niramoy medical chatbot**, ensuring it could process **Bangla medical queries efficiently**.
- Designed the overall **AI pipeline**, refining **attention mechanisms and entropy-based token selection** for better performance.
- **Tanvir Mahtab:** *(Model Evaluation & Testing)*
 - Conducted **comparative testing** between different transformer architectures.
 - Evaluated **model accuracy, efficiency, and robustness**, ensuring optimal performance.
 - Provided **feedback on training iterations** to improve response generation.
- **Mohammad Ashik Jaman:** *(Front-End & UI/UX Developer)*
 - Designed and implemented the **chatbot's user interface (UI)** using **HTML, CSS, and JavaScript**.
 - Ensured a **user-friendly and accessible design**, tailored for **Bangla-speaking individuals**.
 - Developed a **separate medical chatbot interface**, contributing additional UI/UX refinements.
- **Rafiul Islam:** *(Backend & API Integration)*
 - Managed **API calls** to connect the front-end interface with the **EAAT model**.
 - Implemented the **model on Hugging Face**, ensuring smooth deployment and accessibility.
 - Sourced and **preprocessed relevant medical datasets** for fine-tuning the chatbot.
- **Tanvir Ahamad Jibon:** *(Supporting Tasks & Assistance)*
 - Assisted in **minor debugging** of the UI components.
 - Provided **basic testing feedback** for front-end elements.
 - Helped in **documenting API response behaviors** for future improvements.

Chapter 4

Implementation and Results

This chapter is devoted to implementing and evaluating the Niramoy AI Bangla Medical Chatbot. The models were made, tested, and compared using cloud-based software. The testing took place on the Google Colab platform utilizing GPU-powered acceleration. This chapter discusses the environment, methodologies, and observations after conducting experiments.

4.1 Environment Setup

Niramoy AI Bangla Medical Chatbot was developed, implemented, and evaluated on Google Colab Free Tier using GPU. This setup provided resources for easy training and reliable evaluation, optimized for medical queries in Bangla literature. The study compares four key models:

- **Base Transformer Model** (Baseline for comparison)
- **Adaptive Token Ranking and Transformation (ATRT)**
- **Recursive Attention Refinement (RAR)**
- **Entropy-Aware Adaptive Transformer (EAAT)**

4.1.1 Hardware and Software Configurations

To ensure sufficient computational power for training and evaluating our model, the following computational resources were used:

- **Platform:** Google Colab Free (Cloud-based)
- **GPU:** NVIDIA Tesla T4 (CUDA-enabled)
- **RAM:** 16 GB
- **Programming Language:** Python 3.9+

Deep Learning Frameworks and Libraries:

- TensorFlow 2.9 / PyTorch 1.12 (for model training and optimization) [11, 12].
- Hugging Face Transformers (for pre-trained models and NLP processing) [16].
- Scikit-learn, NumPy, Pandas (for data preprocessing and evaluation).
- Matplotlib, Seaborn (for performance visualization).
- NLTK & SentencePiece (for text tokenization and language adaptation) [6, 9].
- TensorFlow Datasets (tfds) (for dataset handling and augmentation).

4.1.2 Dataset and Preprocessing

We trained Transformer architectures on real-world benchmark datasets to build a robust and context-sensitive NLP model.

Datasets Used:

- **AG News Dataset:** A structured text classification dataset categorized into four broad topics.
- **IMDB Dataset:** A binary-labeled sentiment analysis dataset, essential for evaluating classification of subjective queries such as patient-generated medical concerns.

4.1.3 Tokenization Strategy

Different tokenization methods were applied by each Transformer model:

- **Base Transformer Model:** Used a fixed attention mechanism where all tokens in the input sequence were treated with equal weight.
- **Adaptive Token Ranking and Transformation (ATRT):** Dynamically ranked tokens that are contextually significant.
- **Recursive Attention Refinement (RAR):** Applied recursive token filtering, systematically removing redundant inputs.
- **Entropy-Aware Adaptive Transformer (EAAT):** Integrated entropy-driven adaptive attention to adjust attention weights dynamically [1, 8].

Configuration Settings:

- Vocabulary Size: 10,000 tokens
- Maximum Sequence Length: 128 tokens

4.1.4 Training and Hyperparameter Optimization

Pretraining Phase:

- Models were initially trained on AG News and IMDB datasets.
- The goal was to establish a solid understanding of text classification and sentiment analysis.

Optimized Hyperparameters:

- Batch Size: 32
- Learning Rate: $5e^{-5}$
- Optimizer: AdamW
- Dropout Rate: 0.1
- Attention Heads: 12 (dynamically adjusted for EAAT)
- Early Stopping: Implemented to prevent overfitting

4.2 Testing and Evaluation

An evaluation framework was designed to compare the models.

4.2.1 Evaluation Metrics

- Accuracy (%)
- Validation Stability
- Test Loss
- Test Accuracy

4.2.2 Experimental Setup

Each model was trained for 10 epochs using Google Colab Free Tier's GPU acceleration.

The final test accuracy and loss scores were obtained from the AG News and IMDB datasets.

4.3 Results and Discussion

The models were compared based on empirical performance metrics:

Model	Feature	Train Acc. (%)	Val Acc. (%)	Test Acc. (%)	Test Loss
Base Transformer	Standard self-attention	92.30	91.10	91.25	0.1984
ATRT	Dynamic token ranking	96.04	94.50	95.32	0.1524
RAR	Recursive token filtering	95.60	93.80	94.95	0.1603
EAAT	Entropy-based attention	97.32	96.10	96.80	0.1215

Table 4.1: Model Performance Comparison

4.3.1 Key Observations

- **Base Transformer Model:** Used as a benchmark; lacked adaptive token prioritization.
- **ATRT:** Improved accuracy due to token ranking but added additional computation cost.
- **RAR:** Used recursive filtering to provide better validation stability.
- **EAAT:** Achieved the best performance using entropy-based attention tuning, yielding 96.80% accuracy [3, 18].

4.3.2 Final Model Selection

The **Entropy-Aware Adaptive Transformer** was selected as the final model:

- Highest accuracy and lowest test loss.
- Robustness in handling unseen data.

4.4 Summary

This chapter presented an empirical, experiment-driven evaluation of Transformer-based models for Bangla medical chatbot applications. The Base Transformer Model, ATRT, RAR, and EAAT were compared. The **EAAT model was selected as the best choice** due to its superior accuracy, stability, and computational efficiency.

Chapter 5

Standards and Design Constraints

5.1 Compliance with Standards

Ensuring that the Niramoy AI Bangla Medical Chatbot is compatible with relevant standards is required for the preservation of quality, consistency, and end-user trustworthiness. Compliance with these standards also provides a strong foundation for development. This chapter covers the relevant software, hardware, and communication standards that are critical to the chatbot's implementation.

5.1.1 Software Standards

The software for the Niramoy chatbot comprises NLP libraries, machine learning frameworks, and databases used for data storage and retrieval. The principal standards are:

- **IEEE 12207 (Software Life Cycle Processes)**: Defines the software development process in entirety, from requirement-gathering to deployment and support [14].
- **ISO/IEC 9126 (Quality of Software)**: Ensures that the chatbot system is functional, reliable, usable, efficient, portable, and maintainable with ease [21].
- **ISO/IEC 27001 (Management of Information Security)**: Establishes privacy and security procedures related to the processing of sensitive health information [15].

5.1.2 Hardware Standards

Although the majority of operations are performed on a cloud AI service, relevant standards for edge deployment or on-premise server setups include:

- **IEC 62304 (Medical Device Software)**: While the chatbot itself is not a physical medical device, compliance with this standard helps ensure systematic software development [23].

- **RoHS and WEEE Directives:** If Niramoy integrates one or more physical embedded systems, compliance with Restriction of Hazardous Substances (RoHS) and Waste Electrical and Electronic Equipment (WEEE) regulations will ensure environmental safety.

5.1.3 Communication Standards

Since Niramoy transmits health-related data from users to a server, compliance with communication standards is necessary.

- **HL7 (Health Level Seven):** A widely accepted international standard for the exchange of clinical and administrative data [16].
- **FHIR (Fast Healthcare Interoperability Resources):** Ensures interoperability between the chatbot and healthcare providers [16].
- **TLS/SSL Encryption Protocols:** Used to encrypt data transfers and prevent unauthorized access to user information [17].

5.2 Design Constraints

The designers of the Niramoy chatbot must consider various constraints, including economic, environmental, ethical, and social factors. These constraints impact feasibility, implementation, and the social impact of the system.

5.2.1 Economic Constraint

The development of the Bangla Medical AI Chatbot requires significant investment in data collection, model training, infrastructure, and maintenance. A balance between cost and quality must be maintained to ensure sustainable deployment in resource-limited areas [10, 13].

5.2.2 Environmental Constraint

Training a large-scale AI model is energy-intensive and contributes to carbon emissions. Implementing efficient model optimization techniques, such as EAAT’s adaptive attention mechanism, can reduce computational overhead and minimize environmental impact [1, 8].

5.2.3 Ethical Constraint

Handling sensitive medical data requires strict adherence to ethical principles. The chatbot must ensure patient confidentiality and data privacy while preventing the misuse of AI-generated health recommendations. Compliance with ethical guidelines is crucial for gaining public trust [15, 17].

5.2.4 Health and Safety Constraint

Although Niramoy is an informational tool, users may rely on its recommendations to make medical decisions. Therefore, the chatbot must provide medically accurate information and include disclaimers stating that its advice is not a substitute for professional consultation. Collaboration with certified medical professionals can enhance reliability [4, 5].

5.2.5 Social Constraint

Cultural and linguistic considerations play a crucial role in chatbot adoption. A user-friendly interface that respects the nuances of Bangla-speaking communities can facilitate trust and engagement. Ensuring inclusivity in chatbot responses reduces the risk of misinterpretation and enhances accessibility [6, 9].

5.2.6 Political Constraint

Government regulations and healthcare policies can significantly impact chatbot development and deployment. Ensuring compliance with local healthcare laws and data protection policies (similar to HIPAA) is essential for avoiding legal complications and securing regulatory approval [15, 16].

5.2.7 Sustainability

Maintaining up-to-date medical knowledge, research advancements, and treatment guidelines is necessary for chatbot sustainability. Regular updates to the AI model, integration of new medical datasets, and advancements in NLP techniques will ensure long-term viability [11, 12].

5.3 Cost Analysis

The operational costs of the Niramoy chatbot include cloud storage, model training, deployment, and maintenance. Key cost factors include:

- **Cloud Infrastructure:** Covers expenses for CPU, GPU, and storage for model training and inference [16, 19].
- **Data Acquisition:** Includes sourcing, curating, and translating medical corpora into Bangla.
- **Model Maintenance:** Involves continuous supervised training of EAAT and ClinicalBERT with updated medical materials.
- **Personnel:** Covers developers, AI specialists, and medical professionals involved in model validation and supervision.

A continuous cost-benefit analysis is necessary to ensure long-term sustainability and maximize project benefits.

5.4 Complex Engineering Problem

The development of an AI-driven Bangla medical chatbot presents complex engineering challenges that require interdisciplinary collaboration and innovative solutions.

5.4.1 Complex Problem Solving

Some of the key challenges include limited language resources, high accuracy requirements for medical responses, and multi-layered ethical considerations. Solutions to these challenges involve:

- **EAAT:** Optimized algorithms for efficient data processing.
- **Iterative Validation:** Involves continuous review and feedback from medical professionals.
- **Scalability Optimization:** Ensures high-performance AI inference with minimal latency [1, 8].

5.4.2 Engineering Activities

The engineering team responsible for Niramoy must undertake:

- **System Integration:** Configuring NLP models with databases and front-end interfaces.
- **Performance Tuning:** Optimizing the EAAT model for latency, accuracy, and energy efficiency.
- **Quality Assurance:** Conducting rigorous testing for functional correctness, security, and user acceptance.
- **Compliance Management:** Ensuring adherence to medical, legal, and ethical regulations [15, 17].

5.5 Summary

This chapter presented an overview of the standards and design constraints governing the Niramoy AI Bangla Medical Chatbot. Adherence to software, hardware, and communication standards enhances system reliability and security. Economic, ethical, health, and social constraints shape the feasibility and impact of the chatbot. Cost analysis ensures financial sustainability, while complex engineering challenges highlight the technical sophistication required to build an AI-driven medical chatbot for Bangla-speaking users.

Chapter 6

Conclusion

This chapter provides a comprehensive summary of the research, discussing the key achievements, limitations, and directions for future work in developing the Niramoy AI-driven medical chatbot for Bangla-speaking communities.

6.1 Summary

Over the course of this research, Niramoy has evolved from a theoretical concept to an actual AI-driven medical chatbot for Bangla-speaking communities. We identified principal healthcare challenges early: the absence of local language resources, the unequal distribution of certified medical professionals, and the abundance of inaccuracy from naive translation-based methods. We tackled this by developing an Entropy-Aware Adaptive Transformer (EAAT) framework that achieves optimal computational efficiency without compromising on medical inference performance [1].

The cornerstone here was to demonstrate how, with the incorporation of knowledge specific to a field adopted from ClinicalBERT, MIMIC-III, and other multi-domain datasets in Bangla [18], users would be able to hold medical consultations in their local languages without sacrificing precision. Not only was translational fault tolerance provided in the usual way, but even medical vocabulary complexity was addressed such that significant symptoms and associated medical conditions took primacy.

From data pre-processing to deployment, every process was guided with a purpose of augmenting real-world impact. Morphological examination and tokenization ensured fidelity in the treatment of the Bangla text, and named entity recognition (NER) tagged symptoms, diseases, and medications with a special degree of care [3]. Adaptive learning techniques were also adopted in order to balance computational loads, something particularly useful in medical settings where reaction times are paramount. Our modest-scale user tests reported high satisfaction ratings based on improved and faster responses and the system’s ability to handle colloquial expressions in Bangla.

In all, Niramoy is a testament to what is possible in filling a significant healthcare void: getting medical counsel on a timely basis to groups systematically under-served by

mainstream, English-language-based products [5].

6.2 Limitations

Despite the encouraging results, there are a variety of limitations we need to appreciate in having gone ahead with this project. The first is the shortage of high-grade medical datasets in Bangla. We settled on data augmentation with English-to-Bangla translations, but these are not free from subtle inaccuracies [7]. In medical situations in particular, even a minor inaccuracy could taint the chatbot’s suggestions.

Second, our in-the-wild tests were constrained by resources. While initial user tests indicated favorable acceptability, the tests only engaged a relatively small and demographic-limited population. Medical conditions and usage habits are quite variable by geographic area and age and, consequently, broader tests are necessary in order to assure true generalizable performance [9].

A third concern is the long-term consistency of clinical information. Medical knowledge is constantly changing—new research findings and updated clinical practice guidelines are reported with regularity. While we designed in the capability to draw information from recognized sources [4], we require frequent updating and strict validation in order to maintain information in sync with current practice.

Lastly, data privacy is an ongoing concern. Our solution includes provision for encryption and compliance with standards akin to HIPAA [15], but medical information is extremely sensitive. Any broad rollout of Niramoy would have to be accompanied with strict security provisions protecting users’ personal and medical information against access and exploitation [17].

6.3 Future Work

This project momentum paves the way for improvement and expansion in several areas.

6.3.1 Expanded Clinical Dataset Collection

Creating a more comprehensive medical corpus for Bangla, along with verification, would greatly increase chatbot accuracy, and lessen reliance on translation [24]. Joining hands with local hospitals, clinics, and research establishments could result in acquisition of real-world patient data and doctor-patient communication records, thus better domain-specific fine-tuning.

6.3.2 Voice Interaction and Multimodal Support

Many rural users may not know how to read text and would prefer to interact via conversations. If we develop the speech-to-text and text-to-speech modules for various dialects

of Bangla, Niramoy will be usable by more people. Later we can include multimodal data like images [22].

6.3.3 Comprehensive Health Monitoring

Apart from the existing health issue, preventive health can also be integrated. Frequency checks on diet, exercise frequency, or mental wellness may further Niramoy as a health coach towards appropriate lifestyle changes at the individual level [10].

6.3.4 Refinement of EAAT Efficiency

The Entropy-Aware Adaptive Transformer has already put to rest some large-scale transformer inefficiencies. Still, further reducing the model size and improving the hardware-specific optimizations would enable it to run on low-end devices or in bandwidth-constrained scenarios—an important consideration for deployment in rural areas [12].

6.3.5 Integration with Telemedicine Services

A good step forward for Niramoy will be to integrate it with available telemedicine platforms. This can connect users to a video or phone consultation with a certified medical professional from a chatbot consultation [5]. Collaboration would bridge the gap between AI-driven guidance and in-person clinical validation.

6.4 Concluding Remarks

Researchers have used AI that understands natural language to achieve real-world outcomes in healthcare. We have built a unique Transformer that uses domain knowledge and trained it on relevant medical data to help democratize healthcare [11]. We believe that improvements are possible and necessary; nevertheless, it is evident from the results here that Bangla-speaking communities can benefit from AI medical assistance designed with respect to their language and culture.

Niramoy’s future journey remains far from complete looking ahead. As medical science and AI technologies get better, we will have to keep changing the system’s frameworks to integrate new things and learnings coming in [19]. However, the work presented in this thesis can be a very good starting step for anyone who wishes to combine computational linguistics, advanced neural architectures, and health service into a real-world and life-improving solution for the marginalized community.

By considering these changing needs and hurdles, we envisage a future when language will not prevent timely, accurate, and reliable medical advice. This vision can benefit millions of people in Bangladesh and in other low-resource settings around the world [14].

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